

MICHAEL SCHMIDT

Senior Python / Flask Backend Engineer

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PROFESSIONAL SUMMARY

Senior Python / Flask Backend Engineer with 12+ years across API platforms, distributed services, integrations, billing workflows, and enterprise modernization using Python 3.8-3.12, Flask, REST APIs; brings architecture, legacy modernization, secure delivery, CI/CD quality, production troubleshooting, mentoring, and performance optimization to ship maintainable systems for cloud, SaaS, property technology, nonprofit CRM, and civic technology environments.

SKILLS

Backend Engineering: Python 3.8-3.12, Flask, REST APIs, REST APIs, GraphQL, OpenAPI, microservices, background workers, API gateways

Messaging: Kafka, RabbitMQ, SQS/SNS, pub/sub workflows, idempotency, retries, dead-letter queues

Testing & Reliability: unit tests, integration tests, contract tests, load testing, tracing, metrics, structured logging

Data Stores: PostgreSQL, MySQL, SQL Server, Redis, DynamoDB, Elasticsearch, query tuning, migration planning

Cloud & Runtime: AWS, Azure, Docker, Kubernetes, Lambda, ECS, Helm, Terraform, Linux, container security

Senior Practices: system design, production debugging, legacy modernization, mentoring, CI/CD, secure SDLC

WORK EXPERIENCE

Senior Python / Flask Backend Engineer | Lightedge

March 2020 - May 2026

Houston, TX

- Led **backend service architecture** for cloud services and SaaS operations using **Python 3.8-3.12, Flask, REST APIs**, aligning product goals, security requirements, and delivery plans into scalable releases for customer-facing and internal platforms.
- Built and modernized **secure APIs, integration services, background jobs, audit workflows, and operational reporting endpoints** while coordinating architecture, implementation, code reviews, and production support across distributed engineering teams.
- Resolved recurring production issues involving **deadlocks, timeout spikes, queue backlogs, duplicate webhook deliveries, and high-volume database contention**, improving service reliability and reducing user-impacting incidents by **31%** through profiling, safer releases, and stronger observability.
- Modernized **legacy service modules into versioned APIs, stronger database boundaries, and container-friendly deployments**, using practical migration slices, compatibility layers, regression testing, and feature flags to protect revenue workflows during active releases.
- Designed secure integration patterns with **REST APIs, GraphQL, OAuth2/OIDC, JWT, audit logging, and least-privilege access controls** for customer, admin, and partner-facing workflows.
- Improved deployment quality with **Docker, Kubernetes, GitHub Actions, Terraform, automated test gates, and rollback playbooks**, reducing manual release risk across critical services.
- Partnered with product, design, QA, and operations teams to translate ambiguous business needs into technical plans, acceptance criteria, architecture decisions, and measurable delivery outcomes.
- Mentored engineers through design reviews, pairing, refactoring plans, troubleshooting sessions, and coding standards that improved maintainability across shared services and UI modules.
- Strengthened production ownership with **OpenTelemetry, Datadog, CloudWatch, structured logging, SLO reviews, and incident postmortems**, creating clearer accountability for performance and reliability.

Python / Flask Backend Engineer | AppFolio

February 2017 - January 2020

Goleta, CA

- Delivered **backend service architecture** for property technology workflows using period-appropriate versions of **Python 3.8-3.12, Flask, REST APIs**, supporting leasing, payments, reporting, and account-management features.
- Implemented **secure APIs, integration services, background jobs, audit workflows, and operational reporting endpoints** with pragmatic domain modeling, API contracts, validation rules, and reusable components that helped product teams ship complex workflows safely.
- Reduced high-volume workflow latency by **24%** after profiling database queries, API payloads, client rendering paths, cache behavior, and background processing bottlenecks.
- Migrated older modules toward **legacy service modules into versioned APIs, stronger database boundaries, and container-friendly deployments**, balancing modernization with stability for teams maintaining large production codebases and active customer accounts.
- Diagnosed defects related to race conditions, stale UI state, authorization edge cases, and inconsistent validation between clients and services, then added regression tests around the risky paths.
- Built CI/CD improvements with **Jenkins, GitHub workflows, automated test suites, static analysis, and environment configuration checks** to reduce late-cycle release surprises.
- Collaborated with design and product partners to refine flows, reduce support confusion, document edge cases, and keep engineering tradeoffs visible during planning.
- Reviewed pull requests for maintainability, accessibility, security, and performance, helping teammates make changes that were easier to test, release, and support.

Software Developer | Blackbaud

January 2015 - January 2017

Charleston, SC

- Developed software for nonprofit CRM, fundraising, payment, and reporting workflows using mature web and service technologies suited to the **2015-2017** engineering environment.
- Built **secure APIs, integration services, background jobs, audit workflows, and operational reporting endpoints** with careful validation, permissions, export logic, and operational safeguards for teams handling sensitive donor and transaction data.
- Improved critical report generation and workflow response times by **28%** through query tuning, pagination fixes, caching adjustments, and removal of unnecessary service calls.
- Modernized older modules by replacing fragile scripts, duplicated business rules, and tightly coupled service calls with cleaner APIs, shared helpers, and better test coverage.
- Resolved production bugs involving date-range filters, null handling, browser compatibility, batch job failures, and inconsistent data formatting across reporting screens.
- Supported release readiness with smoke tests, rollback notes, deployment checklists, defect triage, and close coordination with QA and support stakeholders.
- Documented technical decisions and mentored junior developers on debugging practices, source control hygiene, testable code, and maintainable feature delivery.

Junior Software Developer | Granicus

June 2013 - December 2014

Dallas, TX

- Built civic technology features for public-sector workflows using practical web, database, and service patterns common to the **2013-2014** delivery environment.

- Implemented account, content, notification, and reporting improvements while learning production engineering practices across requirements, coding, testing, and release support.
- Fixed defects involving form validation, session handling, browser rendering differences, SQL performance, and inconsistent integration responses in public-facing applications.
- Refactored older pages and utilities into cleaner modules with reusable helpers, clearer naming, safer error handling, and better separation between UI and data logic.
- Assisted with database scripts, deployment verification, support investigations, and issue reproduction steps to keep releases predictable for government customers.
- Collaborated with senior engineers during code reviews and troubleshooting sessions, building strong habits around documentation, testing, and maintainable delivery.

SELECTED PROJECTS

- **Integration API Platform:** Built secure partner APIs with **Python 3.8-3.12**, **Flask**, **REST APIs**, OpenAPI contracts, OAuth2/OIDC, queue-backed retries, idempotent webhooks, structured logging, and deployment automation.
- **Legacy Service Upgrade:** Modernized legacy backend modules into container-ready services with safer database migrations, contract tests, tracing, and rollback-friendly release practices.

CERTIFICATIONS

- **AWS Certified Developer - Associate**
- **Microsoft Certified: Azure Developer Associate**
- **Certified Kubernetes Application Developer (CKAD)**
- **Professional Scrum Developer I**

EDUCATION

Texas A&M University-San Antonio - Bachelor's Degree, **Computer Science** (2009 - 2013)